



Semiconductor moiré materials

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Moiré materials have emerged as a platform for exploring the physics of strong electronic correlations and non-trivial band topology. Here we review the recent progress in semiconductor moiré materials, with a particular focus on transition metal dichalcogenides. Following a brief overview of the general features in this class of materials, we discuss recent theoretical and experimental studies on Hubbard physics, Kane–Mele–Hubbard physics and equilibrium moiré excitons. We also comment on the future opportunities and challenges in the studies of transition metal dichalcogenide and other semiconductor moiré materials.

Arguably, one of the most exciting developments in condensed-matter physics in the past few years is the discovery of moiré materials^{1–4}. These materials form when two layers of van der Waals materials stack on top of each other with either a small twist angle or a lattice mismatch. A moiré superlattice structure emerges as a result of the interference in the atomic lattice structures of the constituent materials (Fig. 1a). The superlattice constant or the moiré period a_M is many times bigger than the atomic lattice constant a . This difference in length scales allows the separation of the low-energy physics (set by the scale a_M) from the high-energy physics (set by the scale a)⁵. Electrons in moiré materials can thus be treated as moving in a smooth periodic potential (Fig. 1b); the atomic lattice structure of the constituent materials can be discarded as long as low-energy physics is concerned. Bloch's theorem applies to the mini-Brillouin zone associated with the superlattice and moiré flat bands emerge (Fig. 1c).

The emergence of the new length scale a_M makes moiré materials a highly controllable quantum system. The moiré atomic density of $\sim a_M^{-2}$ is about three orders of magnitude smaller than the real atomic density $\sim a^{-2}$. The maximum doping density in a typical field-effect device before dielectric breakdown is up to about 1% of a^{-2} . Electrostatic gating can therefore fill many electrons in each artificial moiré atom (with size $\sim a_M$; Fig. 1b). This is important because every integer filling of electrons is analogous to a different chemical element in real materials. In addition to controls of electron filling, the moiré flat bands, which have a bandwidth that is about one to two orders of magnitude smaller than that of real materials, are also highly susceptible to external stimulus, such as external electric fields^{4,6,7} and pressure⁸. In general, terms in a many-body moiré Hamiltonian are continuously tunable.

The emergence of the moiré length scale also makes moiré materials a unique platform for studies of strong correlation physics^{9–11}. Because the Coulomb repulsion between two electrons in the same moiré site scales as $U \propto a_M^{-1/2}$ (ref. 12) and the moiré bandwidth (or the kinetic energy in two dimensions for massive electrons) scales as $W \propto a_M^{-2}$, U completely dominates over W for large moiré periods. Electrons in moiré materials with flat bands are strongly correlated and the correlation is tunable by varying a_M .

Experimental studies on moiré materials started soon after the discovery of graphene^{13–15}. Scanning tunnelling microscopy (STM) studies have directly observed the formation of moiré patterns and enhanced electronic density of states in small-angle twisted bilayer graphene¹. Theoretical investigations of the electronic properties

followed almost immediately^{5,16,17}. Intriguingly, moiré flat bands that can substantially enhance electronic correlations were predicted at specific ‘magic’ angles. The magic of these flat-band systems was not revealed until the discovery of superconductivity near correlated insulating states in magic-angle twisted bilayer graphene^{3,18}. The result has attracted tremendous attention in the physics community ever since.

In this Review, we focus on semiconductor moiré materials (readers interested in graphene moiré materials, please refer to refs. 10,11). Compared with twisted bilayer graphene built on Dirac massless electrons (Table 1), semiconductor moiré materials built on massive electrons have two important differences. First, there is no magic-angle condition for the formation of flat bands^{19–21}. Owing to the finite energy gap in two-dimensional (2D) semiconductors, moiré flat bands can be formed in a continuous range of twist angles. Second, the flat bands in semiconductor moiré materials (both topological and non-topological) admit a local tight-binding description (with one or two nearby orbitals) in the moiré lattice^{19,20}. This is, in general, not a good approximation in twisted bilayer graphene²², in which the flat bands form as a result of the interference between intralayer and interlayer hopping at the moiré length scale⁵.

Among all of the 2D semiconductors, we further limit our focus to the family of transition metal dichalcogenide (TMD) semiconductors, in which substantial recent progress has been made. Monolayer TMDs are direct bandgap semiconductors^{23,24} with relatively low disorder densities ($\lesssim 10^{11} \text{ cm}^{-2}$) for electron transport studies²⁵. They have a unique spin–valley-locked electronic band structure and valley-dependent optical selection rules²⁶ that form the basis for optical probes of magnetism. (See refs. 27,28 for a review on the basic properties of monolayer TMDs.) The lattice mismatch in the family of TMDs (which varies from $\sim 0\%$ to $\sim 13\%$) and the tunable twist angle allow the formation of moiré lattices with a wide range of moiré densities ($\sim 10^{12}–10^{13} \text{ cm}^{-2}$). Moreover, heterobilayers built on lattice-mismatched monolayers are generally insensitive to twist-angle disorders compared with homobilayers (Fig. 1d). For these reasons, robust strong correlation and topological physics as well as their interplay have been demonstrated; moiré exciton physics has also been reported. We review these recent developments below.

Moiré superlattice structure

TMD moiré materials form by either stacking two identical monolayers on top of each other with a small twist angle θ

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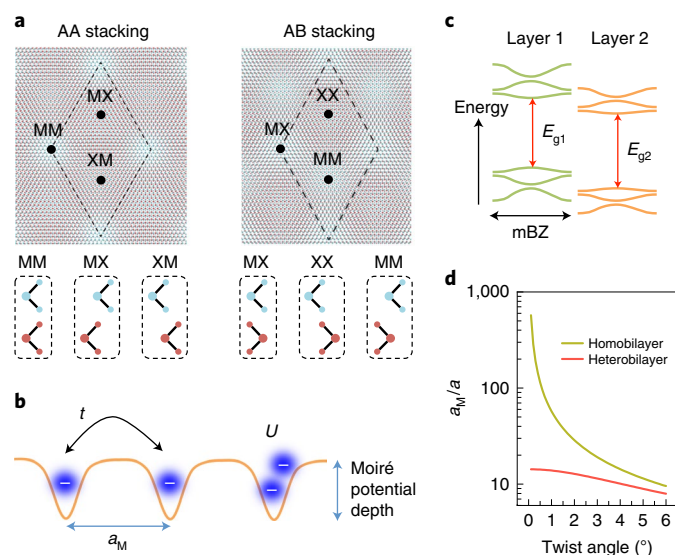


Fig. 1 | TMD semiconductor moiré materials. **a**, Moiré lattice structure for AA-stacked and AB-stacked TMD semiconductors (top). The high-symmetry sites for each case are labelled; their cross-section views are shown (bottom). The large and small dots label the transition metal atom (where M is Mo and W) and the chalcogen atom (where X is S, Se and Te), respectively. **b**, Schematic illustration of an array of moiré atoms that trap electrons, which can tunnel between neighbouring sites with amplitude t and experience on-site Coulomb repulsion U . **c**, Schematic layer-resolved moiré band structure for semiconductor moiré materials with type-II band alignment. mBZ stands for mini-Brillouin zone, and E_{g1} and E_{g2} are the bandgap of the first and second TMD layers, respectively. **d**, Twist-angle dependence of the normalized moiré period a_M/a for both homobilayers and heterobilayers (in the small-angle limit and $\delta = 7\%$). a_M/a is much more sensitive to twist-angle variations in homobilayers.

(homobilayer)^{29–31} or stacking two different monolayers with a lattice mismatch δ (heterobilayer)^{32–37}. A hexagonal superlattice structure is formed with $a_M \approx a/\theta$ for the former and $a_M \approx \frac{a}{\sqrt{\delta^2 + \theta^2}}$ for the

latter (Fig. 1d). The typical moiré period is $a_M \approx 10 \text{ nm} \gg a \approx 1 \text{ Å}$. Owing to the three-fold rotation symmetry in TMD monolayers, there are two different stacking structures: 0° stacking and 180° stacking, in which the two layers are aligned near twist angles of 0° and 180° , respectively. They are also commonly referred to in the literature as AA stacking and AB stacking, respectively (not to be confused with the AA and AB stacking regions in twisted bilayer graphene). We will use this terminology from here on. For AA stacking, there are three high-symmetry sites in the superlattice (Fig. 1a): the MX site, in which the transition metal atom M (Mo or W) lays directly on top of the chalcogen atoms X (S, Se or Te); the XM site, the inverse of the MX site; and the MM (or XX) site with M (X) in one layer directly on top of M (X) in the other. The high-symmetry sites in AB stacking are different (Fig. 1b). These include the MX site (the stacking structure of natural bilayer TMDs) and the MM and XX sites.

The fabrications of homobilayer and heterobilayer moiré structures are quite different. The tear-and-stack technique, commonly used in making twisted bilayer graphene, is employed for homobilayers^{38–40}. A single TMD monolayer obtained from mechanical exfoliation is torn into two halves, which are then re-stacked on top of each other with a controllable twist angle θ . In contrast, two different TMD monolayers obtained from separate exfoliations are involved in the fabrication of heterobilayers. Predetermination of the crystal axis for each monolayer is required to create angle-aligned

Table 1 | Comparison between twisted bilayer graphene and TMD semiconductor moiré materials

Parameter	Twisted bilayer graphene	TMD semiconductor moiré materials
Sensitivity to twist angle	Magic angle(s) for flat bands Sensitive to twist-angle variations	No magic angle in general Homobilayer (heterobilayer): sensitive (insensitive) to twist-angle variations
Local tight-binding description of low-energy physics	Not a good approximation	Good approximation
Electronic degrees of freedom	Spin and valley degenerate	Spin-valley locked Spin or valley degenerate
Typical energy scales	$W \approx 10\text{--}100 \text{ meV}$ $U \approx 20\text{--}40 \text{ meV}$	$W \approx 1\text{--}100 \text{ meV}$ $U \approx 100\text{--}200 \text{ meV}$

heterobilayers. This is often achieved by polarization- and angle-resolved optical second-harmonic-generation spectroscopy^{32–34}. The typical angle-alignment accuracy is about $\pm 0.5^\circ$, which is less accurate compared with the tear-and-stack method. However, this hardly matters because $a_M \approx a/\delta$ for heterobilayers is dominated by the lattice mismatch $\delta \gg \theta$ (Fig. 1d). The insensitivity of the moiré lattice to the twist angle (and therefore twist-angle disorders) is a major advantage of heterobilayers.

The superlattice structure of TMD moiré materials has been characterized by transmission electron microscopy^{34,41}, STM^{30,42}, piezoresponse force microscopy⁴³ and density functional theory (DFT) calculations^{6,42,44}. The structure is far from the idealized rigid structure without lattice distortion in the constituent monolayers. First, there is lattice reconstruction within each monolayer to maximize the area of the most stable stacking structure in each moiré unit cell. This creates a spatially periodic strain modulation within each monolayer, which modulates the TMD band edges and contributes to the total periodic moiré potential⁴². Second, there is unintentional relative strain between the constituent monolayers from fabrications; it distorts the perfect hexagonal superlattice structure. The unintentional strain can vary randomly over the sample and acts as a disorder potential. Third, unintentional twist-angle variations randomly distributed over the sample creates random variations in the moiré period over a length scale that is smooth compared with a_M and acts as another source of disorder potential. This effect is particularly important in homobilayers⁴³, in which a_M is sensitive to θ . Finally, large-scale reconstructions into random patches of the stable stacking structure and/or stripy patterns can occur in samples with a large moiré period, such as homobilayers with a small twist angle⁴³. The entire moiré superlattice structure can be lost in extreme cases.

Semiconductor moiré flat bands

A good starting point to understand the electronic band structure of TMD and other semiconductor moiré materials is the continuum model^{5,19,20}. In the limit $a_M \gg a$, the high-energy physics set by the atomic scale is well separated from the low-energy physics set by the moiré length scale. To a good approximation, electrons in TMD moiré materials can be treated as particles with an effective band mass m of the constituent monolayers moving in a smooth periodic moiré potential V_M (Fig. 1b), giving rise to a moiré Hamiltonian $H_M = \frac{p^2}{2m} + V_M$ and a series of flat bands in the mini-Brillouin zone (Fig. 1c). Here p is the quasi-momentum operator and V_M can be approximated by a Fourier expansion of harmonics associated with different moiré reciprocal lattice vectors^{5,19,20}.

The spatial dependence of V_M can be obtained by calculating the dependence of the band-edge energy on the relative atomic displacement between the two TMD layers using DFT methods^{19,20,45}. As the relative atomic displacement varies smoothly in the moiré length scale, the band-edge modulation corresponds to the moiré potential. The monolayers are treated as rigid in this approximation; only the contribution to V_M from spatially modulated interlayer hopping is taken into account. Another method to calculate V_M is to first obtain the moiré flat bands using large-scale DFT calculations that include lattice relaxations in the superlattice^{6,21,42,44}; the moiré band dispersion is then fit to a continuum model to obtain V_M (refs. 44,46). Contributions to V_M from both spatially modulated interlayer hopping and periodic intralayer strain from lattice relaxations are taken into account in this method. The relaxed structure is, however, sensitive to the choice of van der Waals functionals.

Hubbard model physics

For angle-aligned TMD heterobilayers and twisted homobilayers with layer asymmetry (for example, under an electric field perpendicular to the sample plane), the layer degeneracy of the electronic bands is lifted. The intrinsic Ising spin–orbit coupling in the constituent TMD monolayers also lifts the spin degeneracy within each electronic valley^{26–28}. As a result, only a two-fold valley degree of freedom with approximate SU(2) symmetry remains for the first moiré conduction/valence band^{12,19}. This isolated flat band can be mapped to a tight-binding model in a triangular lattice with a nearest-neighbour hopping term t (ref. 19). The effect of electronic Coulomb interactions can be included as an on-site repulsion U and an extended repulsion V (refs. 19,45,47,48). The system effectively realizes a single-band extended Hubbard model on a triangular lattice^{19,45,47–49} (Fig. 1a,b). For AB-stacked homobilayers, in which interlayer hopping is spin-forbidden, a bilayer Hubbard model with approximate SU(4) symmetry in the valley-layer space can also be realized^{20,50,51}.

The hopping term can be estimated as $t = \frac{W}{9} \approx \frac{\hbar^2 \pi^2}{9ma_M^2} \approx 1 - 5$ meV for a typical band mass $m \approx 0.5m_0$ in TMDs⁵² (m_0 is the free electron mass) and for a typical moiré period $a_M \approx 5 - 10$ nm ($\hbar$ is Planck's constant); the factor 9 comes from the triangular lattice. The on-site repulsion term can be estimated as $U \approx \frac{e^2}{4\pi\epsilon\epsilon_0 a_W} \approx 100 - 200$ meV for a typical size of the localized Wannier orbital $a_W \approx 2$ nm (ref. 19) and a dielectric constant $\epsilon \approx 4 - 5$ of the substrate⁵³ (e and ϵ_0 denote the electron charge and the vacuum permittivity, respectively). The extended repulsion term V can be estimated based on the Coulomb potential and the triangular lattice structure; for instance, the nearest-neighbour Coulomb repulsion is $V_1 \approx \frac{e^2}{4\pi\epsilon\epsilon_0 a_M} \approx 50 - 100$ meV. We can see from these estimates that TMD moiré materials simulate Hubbard model physics in the strong correlation limit $U > V > W$. Experimentally, V can be effectively tuned by the sample–gate distance via screening^{54–56}; t can be tuned by either the sample twist angle^{19,29,45,47} or the electric field that controls the moiré potential depth^{6,7}; the average filling factor ν of electrons/holes per moiré unit cell is tuned by electrostatic gating. Below we summarize recent experimental developments.

Mott insulator. The on-site repulsion term U dominates the physics at $\nu = 1$ (corresponding to half-band filling). In the strong correlation limit $U > W$, a robust insulating state has been observed by multiple experimental groups on various TMD moiré systems (Fig. 2a), which include angle-aligned WSe₂/WS₂ (refs. 32,33), MoSe₂/WS₂ (refs. 54) and MoTe₂/WSe₂ (refs. 6,57), as well as twisted WSe₂ (refs. 7,29,51) and MoSe₂ (ref. 31). The insulating state is not expected from single-particle band theory. It is consistent with the emergence of a Mott insulator⁵⁸ for a sufficiently isolated moiré band or else a charge-transfer insulator¹². The insulating state emerges as

an attempt of the system to minimize the on-site Coulomb repulsion by localization of the charge carriers. The charge degree of freedom is frozen; the low-energy physics is governed by the collective spin excitations from the local magnetic moments⁵⁹. The magnetic ground state is expected to be 120° Néel order⁴⁵ (Fig. 2a). The emergence of local magnetic moments has been verified by temperature-dependent magnetic circular dichroism (MCD) measurements³². Owing to the spin–valley-locked band structure and the valley-dependent optical selection rules in monolayer TMDs^{26–28}, the MCD signal is directly proportional to the sample magnetization. A Curie–Weiss behaviour for the magnetic susceptibility with a negative Curie–Weiss temperature is observed (Fig. 2b). The result is consistent with the Anderson superexchange interaction between the local magnetic moments in a Mott insulator⁵⁹. The weak superexchange energy scale of $\sim 1 - 30$ K (refs. 6,32) is a result of the distant local moments (separated by $a_M \approx 5 - 10$ nm).

Continuous Mott transition. Whereas the system at $\nu = 1$ is a Mott insulator in the $U \gg W$ limit, it is a Fermi liquid in the opposite limit $U \ll W$. A transition between the two states of matter is expected to happen at $U \approx W$ (refs. 58,59). Such a bandwidth-tuned metal–insulator transition is often driven to first order and accompanied by simultaneous magnetic, structural and other forms of phase transitions⁵⁹. However, the geometric frustration in a triangular lattice and the reduced dimensionality in TMD moiré systems can quench these additional phase transitions and favour a continuous transition from a non-magnetic Mott insulator to a Fermi liquid without symmetry breaking, known as the continuous Mott transition^{60–62}. The non-magnetic Mott insulator near the transition has also been predicted to be some form of quantum spin liquid^{60–66}.

The possibility of realizing bandwidth-tuned Mott transitions in TMD moiré materials has been discussed by refs. 67,68. Recent experiments on two different TMD moiré systems, twisted WSe₂ (ref. 7) and AA-stacked MoTe₂/WSe₂ heterobilayers⁶, have reported a continuous metal–insulator transition at a fixed filling factor $\nu = 1$ induced by electric-field tuning of the hopping term t (Fig. 2d,e). In AA-stacked MoTe₂/WSe₂, the application of an electric field shifts the moiré band alignment, modifies the interlayer hopping and the moiré potential depth, and therefore changes the bandwidth W or the interaction strength U/W . The transport data show a continuously vanishing charge gap as the critical point is approached from the insulating side and, from the metallic side, a diverging quasi-particle effective mass accompanied by an abrupt disappearance of the entire electron Fermi surface. MCD measurements further show a smooth evolution of the magnetic susceptibility across the critical point and the absence of long-range magnetic ordering down to $\sim 5\%$ of the Curie–Weiss temperature. Although many of these observations are consistent with the universal critical theory of a continuous Mott transition^{60,69,70}, recent theoretical studies have pointed out the importance of Coulomb disorders near the metal–insulator transition^{71,72}. The effects of disorder and the nature of the magnetic ground state near the transition⁷⁰ and its relevance to quantum spin liquids deserve further investigations. In twisted WSe₂, a linear-in-temperature resistivity and an anomalous Hall response are observed near the critical point. The results have been interpreted as electric-field tuning of the phase of the complex hopping term t , which induces a paramagnetic metal-to-magnetic insulator transition⁷⁰.

Charge-ordered states. Whereas the on-site U term dominates the physics at $\nu = 1$, the extended repulsion V becomes important away from $\nu = 1$. In the strong correlation limit $U > V > W$, the minimization of the extended Coulomb repulsion between electrons drives the spontaneous formation of incompressible charge-ordered states at commensurate fractional filling factors^{47,48,73,74}. An abundance of charge-ordered states have been observed by multiple

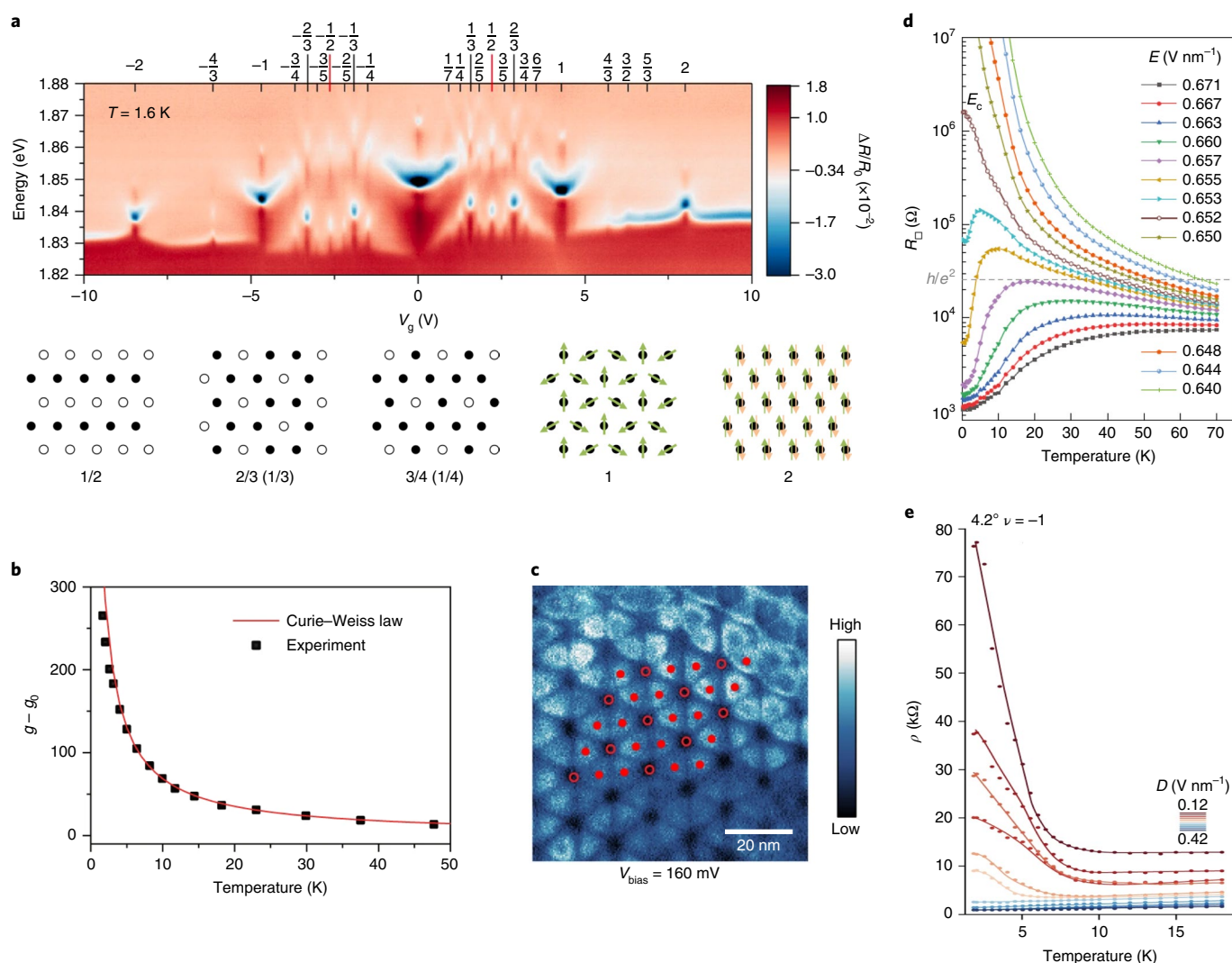


Fig. 2 | Hubbard model physics. **a**, Top: gate-voltage (V_g) dependence of the reflection contrast spectrum ($\Delta R/R_0$) of a Rydberg sensor (monolayer WSe_2) proximal to a WSe_2/WS_2 moiré superlattice. An abundance of insulating states is seen by blueshifts of the $2s$ exciton resonance and enhancements in the spectral weight. The top axis shows the filling factor for the insulating states. Bottom: ground-state charge-order configuration at various filling factors. Filled and unfilled circles denote occupied and empty sites for state ν , respectively. The notation of the occupied and empty sites is switched for the state $1 - \nu$. The ground-state spin configuration (arrows) is also shown for the Mott state and the band insulating state. **b**, Temperature dependence of the moiré exciton g factor (proportional to the spin susceptibility) for the Mott insulating state in a WSe_2/WS_2 moiré superlattice (g_0 is the bare exciton g -factor). The solid line is a Curie–Weiss fit to the data points. **c**, STM image of the charge-order configuration (illustrated by the filled and unfilled circles) for the $\nu = 2/3$ generalized Wigner crystal state in a WSe_2/WS_2 moiré superlattice (V_{bias} is the tip-sample bias voltage). **d, e**, Temperature-dependent resistivity ($R_{\square}$ or ρ) at varying perpendicular electric fields (E or D) applied to $\text{MoTe}_2/\text{WS}_2$ moiré (**d**) and twisted WSe_2 (**e**). A sharp metal–insulator transition is observed at fixed half-band filling. Panels reproduced with permission from: **a**, top, ref. ⁵³, Springer Nature Ltd; **b**, ref. ³², Springer Nature Ltd; **c**, ref. ⁷⁶, Springer Nature Ltd; **d**, ref. ⁶, Springer Nature Ltd; **e**, ref. ⁷, Springer Nature Ltd.

experimental groups using different experimental techniques and on different TMD moiré systems (Fig. 2a), including angle-aligned WSe_2/WS_2 (refs. ^{33,53,75–77}), $\text{MoSe}_2/\text{WS}_2$ (ref. ⁵⁴) and $\text{MoTe}_2/\text{WSe}_2$ (refs. ^{6,57}). The experimental techniques include optical detection of compressibility³³, direct compressibility measurement⁵⁴, Rydberg sensing of dielectric response³³, microwave impedance microscopy⁷⁵ and photoluminescence spectroscopy⁷⁷. In addition, a recent STM study based on imaging the local chemical potential of a graphene gate electrostatically coupled to the charge-ordered states has directly obtained the charge-order patterns⁷⁶ (Fig. 2c). Here the charge-ordered states directly imprint the periodic electrostatic potential onto the nearby graphene gate and modulate the local chemical potential of the graphene.

In addition to charge-ordered states that preserve the rotation symmetry of the system (referred to as generalized Wigner crystals), the competing short-range and long-range Coulomb interactions between electrons can drive the formation of stripe phases that spontaneously break the rotation symmetry^{78–80} (Fig. 2a). The stripe phases can be further divided into stripe crystal phases at commensurate fractional fillings and electronic liquid crystals at incommensurate fillings^{80,81}. The stripe crystal phases are incompressible and have been directly detected by optical birefringence imaging⁸² and STM⁷⁶. The electronic liquid crystals (for example, smectic and nematic phases) are compressible and have been observed by combined optical birefringence and compressibility measurements⁸².

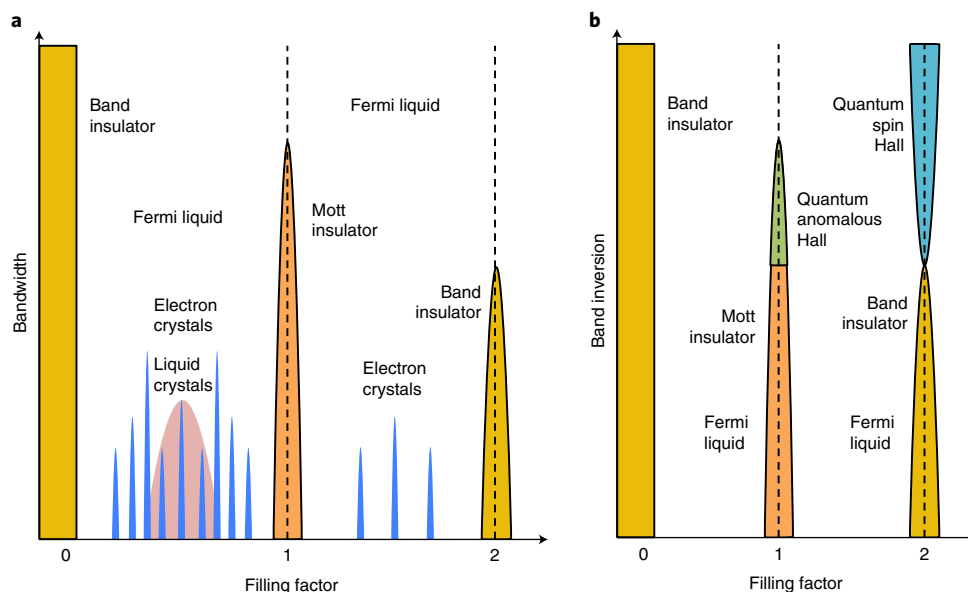


Fig. 3 | Experimental moiré Hubbard and Kane-Mele-Hubbard phase diagrams. **a**, Schematic triangular lattice Hubbard phase diagram (in the bandwidth–filling factor plane) revealed by experiments on various TMD semiconductor moiré materials so far. It is noted that the Wigner transitions at fractional fillings have not been carefully studied. **b**, Schematic Kane-Mele-Hubbard phase diagram (in the band inversion–filling factor plane) revealed by experiments on AB-stacked $\text{MoTe}_2/\text{WSe}_2$ so far. The different insulating states are labelled by different colours.

Finally, we note that similar to the Mott transition, a bandwidth-tuned metal–insulator transition from an incompressible charge-ordered state to a Fermi liquid at fixed commensurate fractional filling is expected at $V \approx W$ (refs. ^{66,83,84}). This is related to the Wigner transition, that is, quantum melting of a Wigner crystal to a Fermi liquid⁸⁵. Dielectric signatures of such a transition have been reported recently⁸⁶. More in-depth studies are required to understand the nature of this quantum phase transition. We summarize in Fig. 3a the experimental Hubbard phase diagram obtained from TMD moiré materials so far.

Kane-Mele-Hubbard physics

AA-stacked homobilayers. Although the Hubbard physics discussed above comes from an isolated, non-topological moiré band¹⁹, intertwining two TMD moiré bands through complex hopping can introduce non-trivial band topology²⁰. This was first predicted in AA-stacked twisted TMD homobilayers²⁰, in which spin-preserved interlayer hopping is allowed. The moiré Hamiltonian has two components, one for each monolayer; it contains the kinetic energy and the intralayer moiré potential for each layer as well as a complex interlayer moiré potential that can intertwine the moiré bands from the two layers. The interlayer hopping splits the otherwise layer-degenerate moiré valence bands into topological bands with non-zero valley-resolved Chern numbers (Fig. 4a,b). For small twist angles, the first two topological moiré bands can be mapped to a Kane-Mele model on a honeycomb lattice⁸⁷ (Fig. 4a), which includes a nearest-neighbour interlayer hopping between the non-equivalent MX and XM sublattice sites, a next-nearest-neighbour intralayer hopping within each sublattice, and an interlayer sublattice potential difference continuously tunable by a vertical electric field. The next-nearest-neighbour hopping term also carries a bond- and spin-dependent phase shift that gives rise to an effective Ising spin-orbit coupling. A transition to non-topological bands is expected at sufficiently large electric fields^{20,46}.

Including the on-site repulsion U in the honeycomb lattice further gives the Kane-Mele-Hubbard Hamiltonian⁸⁸. The presence of both electronic correlation and non-trivial band topology in this model is expected to support a wealth of exotic electronic states of

matter⁸⁹. Depending on the relative strength of the different terms in the Hamiltonian, quantum spin Hall insulator, antiferromagnetic insulator, antiferromagnetic Chern insulator and quantum spin liquids have been predicted at $\nu=2$ (ref. ⁸⁸); Mott insulator, intervalley coherent state and the quantum anomalous Hall (QAH) effect can be stabilized at $\nu=1$ (refs. ^{20,46,90}); the model could also support flat bands with uniform momentum distribution of Berry curvatures that stabilize fractional QAH states⁹¹. The understanding of the complex phase diagram is far from complete. Experimental realization of the model is of considerable interest.

AB-stacked heterobilayers. Surprisingly, the first experimental realization of Kane-Mele-Hubbard physics did not happen in AA-stacked TMD homobilayers near zero perpendicular electric field, as proposed by theory, but in rather opposite experimental conditions: AB-stacked $\text{MoTe}_2/\text{WSe}_2$ heterobilayers under a large electric field⁸⁷. In particular, evidence of an electric-field-tuned continuous topological phase transition from a moiré band insulator to a quantum spin Hall insulator is observed at $\nu=2$, demonstrating the emergence of topological moiré bands (Fig. 4c). At $\nu=1$, a robust QAH state that exhibits zero-magnetic-field quantized Hall transport up to ~ 2 K emerges (Fig. 4e,f); the temperature scale is comparable to that in twisted bilayer graphene^{92,93}. An electric-field-tuned topological phase transition from a Mott insulator to the QAH insulator without charge gap closure is also observed (Fig. 4d). Such a transition is in contrast to the better-known continuous topological phase transitions that involve charge gap closure at the critical point⁹⁴ (for example, that at $\nu=2$). The absence of charge gap closure is probably connected to the different symmetries of the Mott and the QAH insulator⁹⁵. Recent theoretical studies have suggested that the Mott-to-QAH transition is first-order^{46,90,96} and could be of excitonic origin⁹⁷; and the QAH state is valley-polarized^{46,90,96,97}. More in-depth studies are required to better understand the nature of the QAH state and the Mott-to-QAH transition. The experimental Kane-Mele-Hubbard phase diagram mapped so far is summarized in Fig. 3b.

Large-scale DFT calculations have been performed to understand the emergence of topological bands in AB-stacked

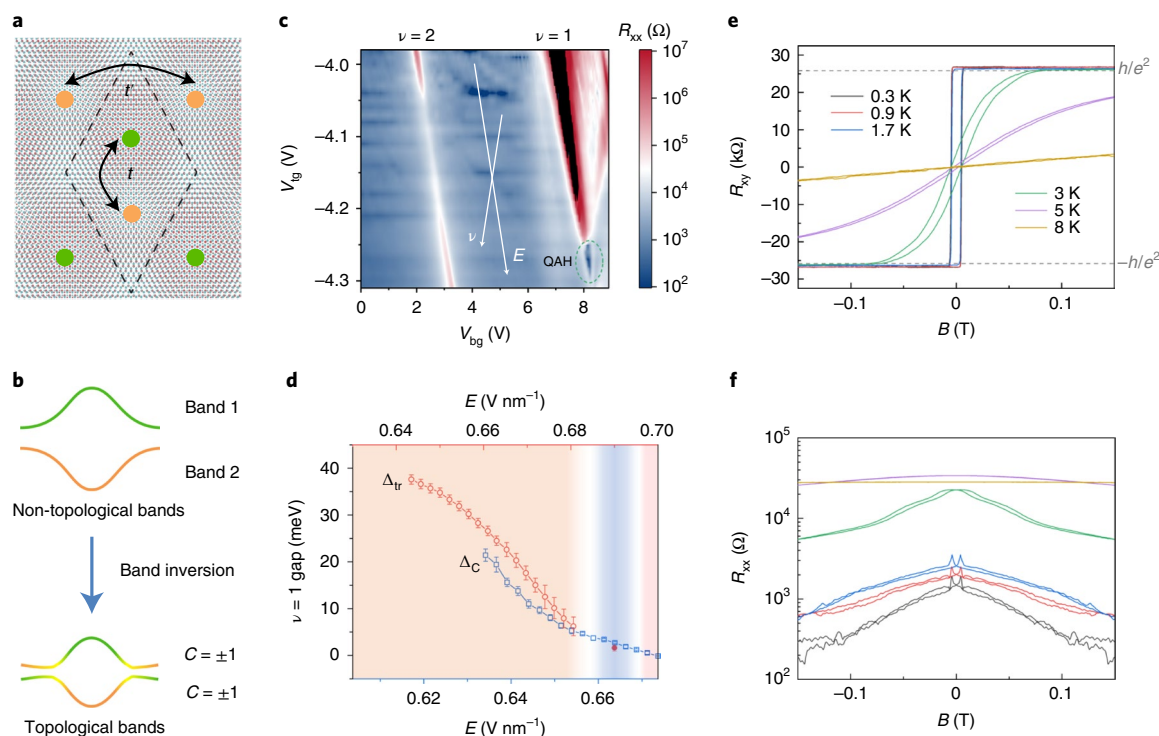


Fig. 4 | Kane-Mele-Hubbard physics. **a**, Illustration of the Kane-Mele model in a honeycomb moiré lattice. The orange and green circles denote the two sublattice sites; the two sites are in general at different electrostatic potentials tunable by a perpendicular electric field. t and t' denote the nearest- and next-nearest-neighbour hopping, respectively. **b**, Schematic illustration of intertwining two moiré flat bands by band inversion and interlayer hybridization. The bands become topological with finite valley-resolved Chern numbers (C) after band inversion. **c**, Dependence of the 300 mK longitudinal resistance (R_{xx}) in AB-stacked $\text{MoTe}_2/\text{WSe}_2$ on the top (V_{tg}) and bottom (V_{bg}) gate voltages. The electric field and filling factor directions are labelled by the arrows. The QAH region with vanishing resistance is circled by the green dashed line. **d**, Electric-field dependence of the charge gap (Δ_C and Δ_{tr}) at half-band filling from the Mott insulating region (orange shaded) to the QAH region (blue shaded) and the metallic region (pink shaded). The red (Δ_{tr}) and blue (Δ_C) data points are obtained by thermal activation fits to the resistance data and by direct compressibility measurements, respectively. Error bars represent the experimental uncertainties from these analyses. **e, f**, Magnetic field (B) dependence of the Hall (R_{xy} , **e**) and longitudinal (R_{xx} , **f**) resistances at varying temperatures for the QAH state. Nearly quantized Hall resistance and vanishing longitudinal resistance are observed at low temperatures. Panels **c–f** reproduced with permission from ref. ⁵⁷, Springer Nature Ltd.

TMD heterobilayers⁴⁴. A tight-binding construction has also been reported⁹⁸. In this theory, the envelope wave function for the first and second moiré valence bands in $\text{MoTe}_2/\text{WSe}_2$ is from the MoTe_2 and the WSe_2 layers, respectively⁴⁴. The two wave functions are centred at the MM and XX sites of the AB-stacked structure, forming a honeycomb lattice. An electric field that tunes the interlayer band alignment causes band inversion. Interlayer tunnelling is allowed because of lattice corrugation in the moiré structure; a gap is reopened after band inversion and topological bands emerge. The Kane-Mele model largely captures the low-energy physics.

Moiré excitons

In addition to fermionic flat bands, bosonic flat bands can also be realized in TMD moiré materials. Because electrons in the conduction and valence bands in general experience different moiré potentials (in both depth and phase), a spatially periodic bandgap modulation arises (Fig. 5a). In the limit that the moiré period is large compared with the exciton Bohr radius, the periodic bandgap modulation dominates the contribution to the moiré potential for excitons (bound electron-hole pairs), and gives excitonic moiré bands^{99,100} (Fig. 5a). The emergence of excitonic moiré bands modifies the optical selection rules^{99–101}. In particular, umklapp scattering off the superlattice provides additional momentum to activate the otherwise momentum-dark excitons in monolayer TMDs^{99–102} (Fig. 5a). Compared with a single dominant excitonic resonance in

the optical absorption spectrum of monolayer TMDs^{27,52}, multiple moiré exciton resonances are expected in TMD moiré materials^{99,101}. Although umklapp scattering induces additional excitonic resonances, the valley-dependent optical selection rule remains valid because the valley pseudospin remains as a good quantum number for $a_M \gg a$ (refs. ^{99–101}). In the following, we review recent progress on the experimental studies of moiré excitons, with a particular focus on equilibrium exciton fluid in a moiré lattice. We refer readers interested in optically excited excitons to recent review articles^{103,104}.

Optically excited moiré excitons. Experimental studies based on optical absorption and radiative recombination of excitons in TMD moiré materials have demonstrated the emergence of excitonic moiré bands^{34–37}. In particular, recent experiments on various TMD combinations (for example WSe_2/WS_2 , $\text{MoSe}_2/\text{WS}_2$, $\text{MoSe}_2/\text{WSe}_2$ and so on) have reported intralayer moiré excitons, which involve transitions between excitonic moiré bands within a single TMD layer^{34,35} (Fig. 5b); interlayer moiré excitons^{36,37}, which involve transitions between states from two different TMD layers; layer-hybridized moiré excitons^{35,105}, which involve transitions between layer-hybridized electronic states; moiré exciton-polaritons¹⁰⁶, strongly coupled moiré excitons and photons; and charged moiré excitons^{107–110}, which are bound states of moiré excitons with electron/holes. Control of the layer-hybridized moiré excitons by the quantum-confined Stark effect has also been demonstrated^{31,105}.

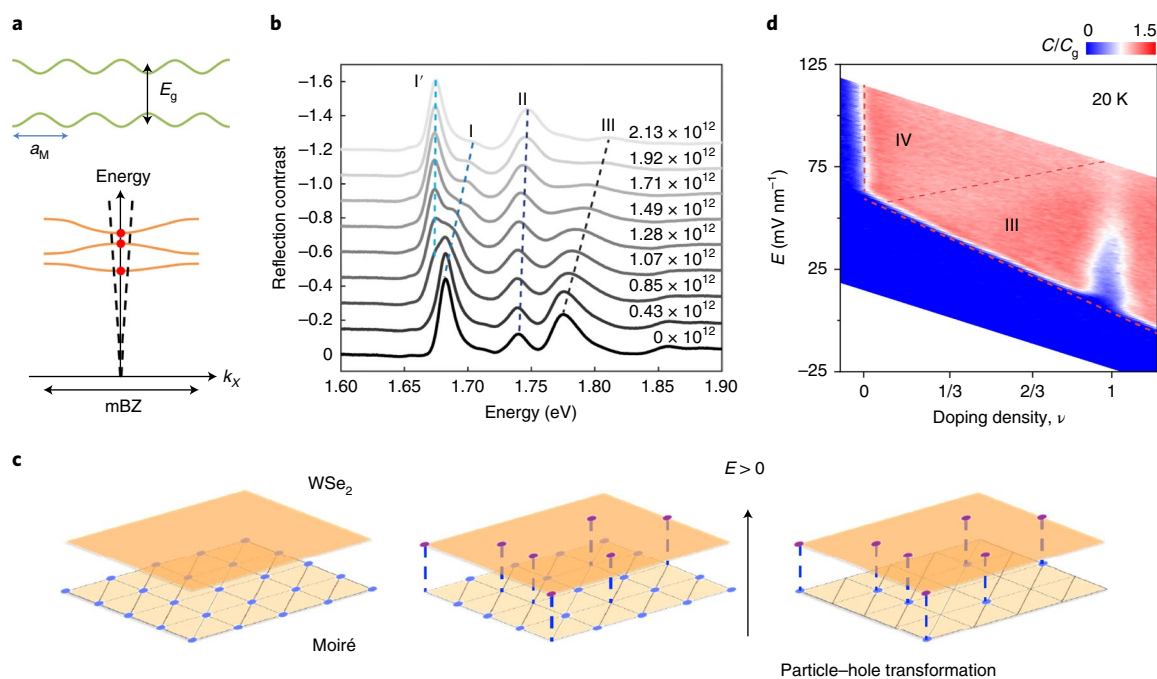


Fig. 5 | Excitons in a moiré lattice. **a**, Schematic illustration of bandgap modulation at moiré length scales (top) and the formation of exciton flat bands in the mini-Brillouin zone (bottom). The light cone is indicated by the dashed lines and the momentum-allowed excitonic transitions near the mini-Brillouin zone centre are denoted by the red dots (k_x denotes the exciton momentum). **b**, Reflection contrast spectrum of WSe₂/WS₂ moiré at various electron-doping densities (in units of cm⁻²). The spectra are vertically shifted for clarity. The exciton replicas (I, I', II and III) are the intralayer moiré exciton resonances in the WSe₂ layer. **c**, Left: a Mott insulating state at half-band filling when all electrons reside in the moiré layer. Middle: the electrons transferred to the TMD monolayer under an electric field are tightly bound to the empty moiré sites to minimize the total Coulomb repulsion energy. Right: under a particle-hole transformation, the empty sites in the moiré layer become holes; a dipolar exciton fluid emerges in the moiré lattice. **d**, Penetration capacitance (C) normalized by the geometrical capacitance (C_g) as a function of the perpendicular electric field and the total filling factor in the Coulomb-coupled WSe₂/WS₂ moiré-WSe₂ monolayer system. Both the moiré layer and the monolayer are hole-doped in region III. An incompressible state is observed at total filling factor 1, corresponding to an excitonic insulating state. Panel reproduced with permission from: **b**, ref. ³⁴, Springer Nature Ltd; **c,d**, ref. ¹¹¹, Springer Nature Ltd.

Moreover, two different types of behaviour have been reported in photoluminescence studies. In twisted MoSe₂/WSe₂, interlayer excitons trapped by disordered moiré potentials, giving rise to a series of sharp photoluminescence peaks, have been observed at low optical pumping intensity^{37,107–110}. At high pumping intensity, the disordered moiré sites with a random distribution of trapping depths become increasingly populated with excitons, leading to a smooth photoluminescence peak that corresponds to a continuous distribution of the defect emissions¹⁰⁹. This behaviour is in contrast to that in angle-aligned WSe₂/WS₂, in which only a smooth interlayer exciton photoluminescence peak is observed down to the lowest optical pumping intensity³². The origin of the different behaviours in the two materials deserves further investigation.

Equilibrium moiré excitons. In addition to non-equilibrium moiré excitons created by optical pumping, equilibrium moiré excitons with density determined by the electrostatics in a capacitor device (rather than by pumping) have also been reported by recent experiments^{111,112}. Such equilibrium exciton fluid in a moiré lattice has the potential to realize the many exotic bosonic phases of matter (for example, bosonic Mott insulators, Wigner crystals, superfluids, supersolids and so on) predicted in the Bose–Hubbard model^{113,114} (Hubbard model for bosons) and the SU(4) Hubbard model⁵⁰ (two-orbital Fermi–Hubbard model). Equilibrium moiré excitons can be realized in either a Coulomb-coupled double moiré system⁵⁰ or a Coulomb-coupled moiré-monolayer system^{111,112}. The spatially separated electrons and holes in these bilayer structures form

interlayer excitons in a lattice (Fig. 5c). To quench the single-particle tunnelling between the constituent layers, they are separated by a thin tunnel barrier; the Coulomb coupling dominates the single-particle tunnelling. In the limit that the barrier thickness is small compared with the moiré period, the interlayer Coulomb correlation is as strong as the intralayer counterpart. The Coulomb-coupled bilayer can be contacted by a single common electrode or by separate electrodes to the electron and hole layer; the electrodes act as an electrical reservoir for excitons^{115–117}. This reservoir allows the top and bottom gates, which can separately control the total doping density (or total filling factor) and the electric field perpendicular to the sample plane, to continuously tune the exciton density and chemical potential. An equilibrium exciton fluid with density determined by electrostatics is realized.

There are two ways to create an equilibrium exciton fluid in the Coulomb-coupled bilayer. The first is to create an electron–hole double layer by electron doping one layer and hole doping the other¹¹⁶. Although this has been realized in Coulomb-coupled TMD double layers without a moiré lattice, its realization in a moiré lattice remains to be demonstrated. The second method relies on a particle–hole transformation^{111,112,118} (Fig. 5c). Consider a Coulomb-coupled moiré-monolayer system with the moiré layer doped at filling factor $\nu=1$ (one electron or hole per moiré site) and the monolayer charge neutral; the entire system is at total filling factor 1. This configuration can be readily realized with the two gates and the electrical reservoir. Under a perpendicular electric field, some charges ($\delta\nu$) can be transferred from the moiré layer

to the monolayer. The moiré lattice allows us to view the system (ignoring spins) as filled with $1 - \delta\nu$ particles in the moiré layer and $\delta\nu$ particles in the monolayer or as $\delta\nu$ holes (or empty moiré sites) in the moiré layer and $\delta\nu$ particles in the monolayer. The particles and holes in the latter particle-hole-transformed configuration are Coulomb bound to minimize the interlayer Coulomb repulsions; an equilibrium exciton fluid in a lattice is formed. This concept has been recently demonstrated by two independent experiments using a Coulomb-coupled WSe_2/WS_2 moiré layer and a WSe_2 monolayer^{111,112}. Both optical and capacitance measurements have shown the emergence of a charge-incompressible state at total filling factor 1 when the system is filled with an exciton fluid; the state is consistent with an excitonic insulating state or an exciton-doped Mott insulator¹¹⁹ (Fig. 5d). The dipolar excitons are expected to be strongly correlated: the dipole-dipole repulsion energy is much larger than the exciton kinetic energy. The results pave the path to explore the Bose-Hubbard^{113,114} and the $\text{SU}(4)$ Hubbard^{50,119} phase diagrams in moiré materials.

Outlook

We believe it is only the beginning of semiconductor moiré materials. There are plenty of opportunities and challenges for both TMD-based and beyond-TMD semiconductor moiré systems. Within the TMD moiré family, much of the Hubbard and Kane-Mele-Hubbard phase diagrams remains unexplored. Outstanding problems include, for example, the nature of the bandwidth-tuned Mott and Wigner transitions; spin liquid physics near these transitions; the existence of superconductivity near the Mott insulator; exotic excitonic phases of matter; fractional QAH physics; and many more. The ability to continuously tune the filling factor, the terms in the many-body Hamiltonian and the moiré lattice symmetry using states from the Γ valley^{73,120} will probably allow the exploration of these problems in new ways. In addition to Hubbard physics, recent theoretical studies have also proposed the realization of Kondo lattice physics in moiré systems with an exchange-coupled itinerant band and a local moment flat band^{121,122}, as well as layer pseudospin liquid physics in Coulomb-coupled moiré double layers^{50,123}. These interesting structures remain to be explored.

Looking beyond TMD semiconductors, there is a large family of 2D semiconductors with different properties and crystal symmetries. These include, for instance, black phosphorus with a buckled in-plane anisotropic structure¹²⁴, III-VI semiconductors with band edges near the Γ point of the Brillouin zone¹²⁵, and a large family of layered magnetic semiconductors with different crystal and magnetic structures^{126,127}. This large set of building blocks will allow us to explore moiré superlattices with symmetries different from the triangular lattice, for example, distorted triangular lattice, square lattice and quasi-one-dimensional lattice. The family of magnetic semiconductors is also an attractive platform to explore moiré magnetism. Recent theoretical studies that made use of the stacking-structure-dependent interlayer exchange interaction in layered magnetic materials have predicted a suite of exotic magnetic ground states in twisted magnetic materials^{128–130}, such as a non-collinear magnetic ground state, a skyrmion lattice and a multi-flavour magnetic state. These states can support novel moiré magnon bands and complex magnon networks¹³¹. Recent experiments have demonstrated the coexistence of ferromagnetic and antiferromagnetic states in small-angle twisted bilayer CrI_3 , providing evidence of a non-collinear magnetic state^{132–134}. These results demonstrate the great potential of magnetic moiré semiconductors in engineering magnetism in the nanoscale.

Outstanding promises are accompanied by outstanding challenges. One of the most important and longstanding challenges in the field of 2D semiconductor materials is the formation of good electrical contacts for transport and thermodynamics studies. Substantial progress has been made in recent years but further

improvements are required. Another important challenge is material quality. In particular, the effects of disorder broadening of the electronic energy levels are much more important in moiré materials with flat bands. Coherent band physics becomes relevant only when such broadening is small compared with the moiré bandwidth. Improvements in both the semiconductor quality and the homogeneity of the moiré superlattice structure are required. Over the years, the material quality of bulk and monolayer TMD crystals has been dramatically improved¹³⁵; improvements in other 2D semiconductor materials will substantially expand the list of interesting semiconductor moiré materials. In terms of improving the homogeneity of the moiré superlattice structure, we believe that thermal-annealing-induced structural relaxation is a promising route in heterobilayers with substantial lattice mismatch because the angle-aligned structure is the lowest-energy structure. Systematic studies are required.

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Competing interests

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